Recognition of Real-Time Indian Sign Language using Hybrid CNN-RNN architecture for help in Critical Scenarios

Presentation by:

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Guided by:

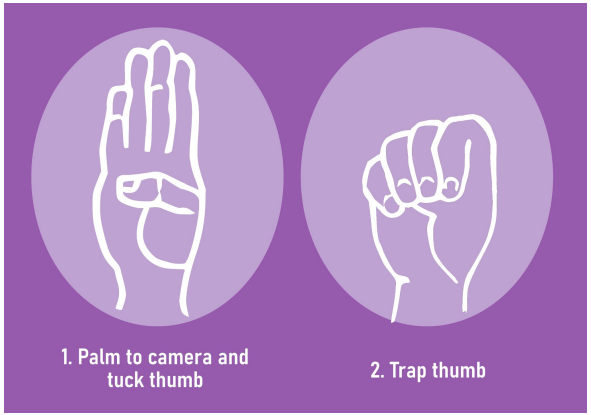
- Dr. J. K Josephine Julina



Introduction







- Palestine - Israel war
- Visual Learners
- Women Safety Sign

Problem Statement

To develop a robust **Indian Sign Language** Detection System focused on **emergency** actions to enhance accessibility and communication for individuals with **hearing** and speaking **impairments** in **critical situations.**



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Literature Survey - Existing Work

NAME	METHODS USED	Features and Limitations
[1] Indian sign language recognition using neural networks and kNN classifiers - 2014	- Hierarchical centroid techniques - kNN - NPR tool	○ Detection of numericals in ISL. ○ Database containing 5000 numeric sign images,with 500 images per numeral sign was created kNN clustering algorithm for classification and recognition of numeric signs
[4] Indian Sign Language Character Recognition - 2015	- HSV model - YIQ and YUV model - Linear kernel SVM - RBF kernel SVM - Random forests	○ Detection of alphabetic characters in ISL. ○ Converted images from RGB Scale to YIQ and YUV space for skin segmentation. ○ Compared different ML models on extracted features and reported accuracy none of which was > 70%.

NAME	METHODS USED	Features and Limitations
[2] Real-Time Recognition of Indian Sign Language - 2019	- Fuzzy c-means clustering - Edge detection - Background noise removal - Morphological transform	○ Significance of facial recognition and hand gestures in ISL ○ Region of Interest identification done using skin segmentation feature of OpenCV ○ Fuzzy C - means clustering algorithm for gesture prediction
[3] Indian Sign Language Recognition using Convolutional Neural Network - 2021	- Conversion to gray-scale - Dilation - Mask operation - Morphological transform - CNN	○ Recognize alphabets from the ISL only ○ Capture images of hand gesture through a web camera ○ Computer vision techniques such as grayscale conversion, dilation and mask operations ○ CNN for model training
[7] Recognition of Indian Sign Language using Machine Learning Algorithms - 2021	- Image processing and cleaning - Comparison of different DL models such as SVM, DTs	○ Detection of ISL alphabets using machine learning algorithms ○ Random Forest - highest accuracy for images

NAME	METHODS USED	Features and Limitations
[6] A comprehensive evaluation of deep models and optimizers for Indian sign language recognition - 2021	- Comparison of different optimizers and pre trained DL models - 3 layered CNN	○ Performed comparison on pretrained ML/DL models such as InceptionV3, ResNet152V2, InceptionResNetV2. ○ Dataset consisting of 716 images for alphabets. ○ Proposed 3 Layered CNN Model - therefore works only on images.
[5] Indian Sign Language Detection using Inceptionv3 Model - 2023	- InceptionV3 model	○ Identification of hand gestures using Images ○ Use of Inceptionv3 model architecture for predicting the images to the corresponding classes
[8] Recognition of Indian Sign Language (ISL) using deep learning model - 2023	- CNN-DSC - Input with glove sensor	○ Dataset comprises 5408 static images (aim to make the system more robust) ○ Uses CNN using 'Depthwise Separable Convolution' - for identifying two hand gestures

Gaps Identified

- Detection of only numericals, characters and alphabets
- Detection and Recognition from Simulation and not Real Time.
- Sequence and temporal features not detected.
- Classification or Identification from images and not video datasets.
- All systems are currently not trained and tested completely in real-world scenarios.

Scope & Feasibility

- Individualized Gesture Recognition - Adapting to the changes in the way each person performs the sign language act.
- Real time processing - Immediate recognition of the Sign Language acts
- Spatial and temporal modeling.
- Feedback Mechanism
- Data Availability
- Computational Resources

Objectives

- Identify different signs accurately, specifically targeting seven classes: "alive," "dead," "mother," "fast," "loud," "healthy," and "sick."
- Utilize MediaPipe for real-time hand motion tracking.
- Implement a hybrid CNN-RNN architecture for spatial and temporal data analysis.
- Enable the system to categorize and recognize sign language without lag and with high accuracy.
- Ensure adaptability to varying signing styles and gestures.
- Incorporate feedback mechanisms for continuous improvement based on real-world usage.

Proposed Solution of the Idea

A combination of **CNN** (Convolutional Neural Network) and **RNN** (Recurrent Neural Networks) is a powerful architecture commonly used for tasks that involve both **spatial and temporal information**. This combination is well-suited for tasks such as video analysis, where CNNs can capture spatial features from individual frames, and RNNs can model the temporal dependencies between these frames.

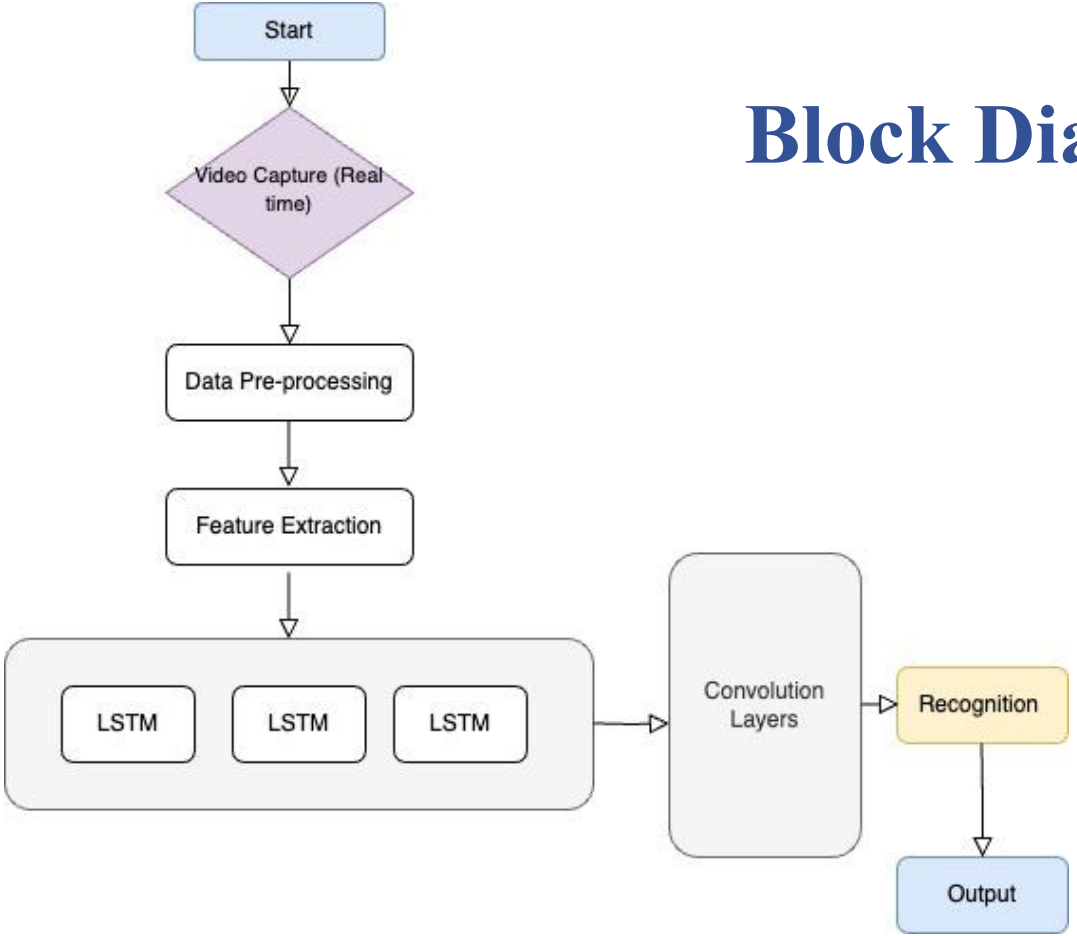
Proposed Solution of the Idea

CNN + RNN offers a hierarchical approach to feature learning. CNNs capture spatial features hierarchically from low-level to high-level, and LSTMs model temporal dependencies in a sequential manner.

CNN for - Spatial Information
Effective at capturing spatial features from images, which is crucial for understanding hand gestures and movements in sign language.

RNN for - Temporal Modeling
Capable of capturing long-term dependencies and sequential patterns in the data. It can handle variations in the duration and timing of gestures, making the model more robust to different signing styles.

Block Diagram



Dataset Description

We are using Indian Lexicon Sign Language Dataset (**INCLUDE**) as our primary dataset.

INCLUDE fills a significant void in Sign Language Recognition (SLR) research, serving as the first publicly available dataset for ISL.

INCLUDE has been recorded under the guidance of experienced signers and with input from the **deaf community and reflects authentic communication scenarios**, facilitating the advancement of ISL recognition technology.

The main advantages of this dataset is that it is **open sourced** and it imitates the **real-world scenarios**.

Dataset Description

INCLUDE comprises **4,287 videos spanning 263 classes**, where each class represents a distinct ISL sign.

These classes are categorized into **15 broad categories**, encompassing commonly used words in ISL, including adjectives, animals, greetings, occupations, and more.

The dataset is characterized by **short video durations**, primarily ranging from 2 to 4 seconds, with an average duration of 2.57 seconds per sign. Each video maintains a resolution of 1920x1080 pixels and a frame rate of 25 fps, resulting in a total of 0.27 million frames encapsulating 263 unique signs.

Dataset Description

From this dataset, words related to **emergency situations** have been **manually picked** to ensure that communication for individuals with hearing and speaking impairments are enhanced, especially in critical situations.

Additionally, the dataset was **manually labeled** for training purposes. This manual labeling ensures that the dataset is reliable and precise, making it valuable for developing systems that recognize ISL in emergency situations.

By adhering to this approach, we aim to train our model on a substantial portion of the data while reserving a **separate subset for unbiased evaluation**, thereby ensuring the reliability and generalization of our real-time recognition system.

Implementation

1. Data Preprocessing and Feature Extraction: Sign language videos undergo preprocessing to ensure compatibility with the neural network model. Spatial features from frames are extracted, capturing crucial movement.

2. Temporal Modeling with RNNs: Sequential data handling, capturing temporal dependencies between frames. They understand the temporal flow of gestures, adapting to variations in timing and style.

3. Integration of CNN and RNN Architectures: CNNs extract spatial features, while RNNs model temporal dependencies. This hybrid approach enables accurate real-time recognition of Indian Sign Language gestures in emergency scenarios.

CNN Implementation Demo

```
 /FYP/code2/test.py"
2024-02-09 22:56:23.669115: I tensorflow/core/platform/cpu_feature_guard.cc:182] This TensorFlow
 binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: AVX2 AVX512F AVX512_VNNI FMA, in other operations, rebuild
 TensorFlow with the appropriate compiler flags.
/Users/pavithramanikandan/Library/Python/3.9/lib/python/site-packages/urllib3/__init__.py:34: No
tOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled
 with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
1/1 [==============================] - 0s 221ms/step
7
7
Predicted category:  quiet
pavithramanikandan@Pavithras-MacBook-Pro code2 % /usr/bin/python3 "/Users/pavithramanikandan/SSN
```

CNN Implementation Demo

- The CNN model achieved an **accuracy of 0.63**, suggesting moderate performance in ISL recognition.

- However, with a **loss of 0.66**, indicating relatively high error, transition to the RNN+CNN implementation was made as the accuracy and robustness was to be improved.

CNN+RNN Implementation



CNN+RNN Implementation



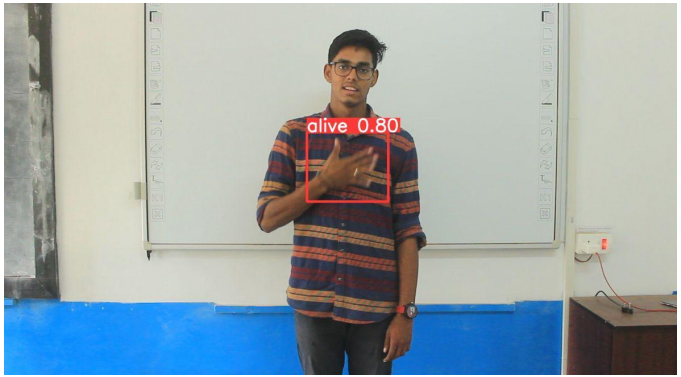
CNN Implementation Demo

- The RNN+CNN model exhibited significant improvement over the previous model, achieving an **accuracy of 0.8** and reducing the **loss to 0.42**.

- The model's higher accuracy and reduced loss signify enhanced precision and robustness, highlighting its potential to foster more seamless communication and inclusivity for individuals who are deaf or hard of hearing.

- To improve the accuracy further, the Mediapipe framework was used.

YOLOv8 Implementation



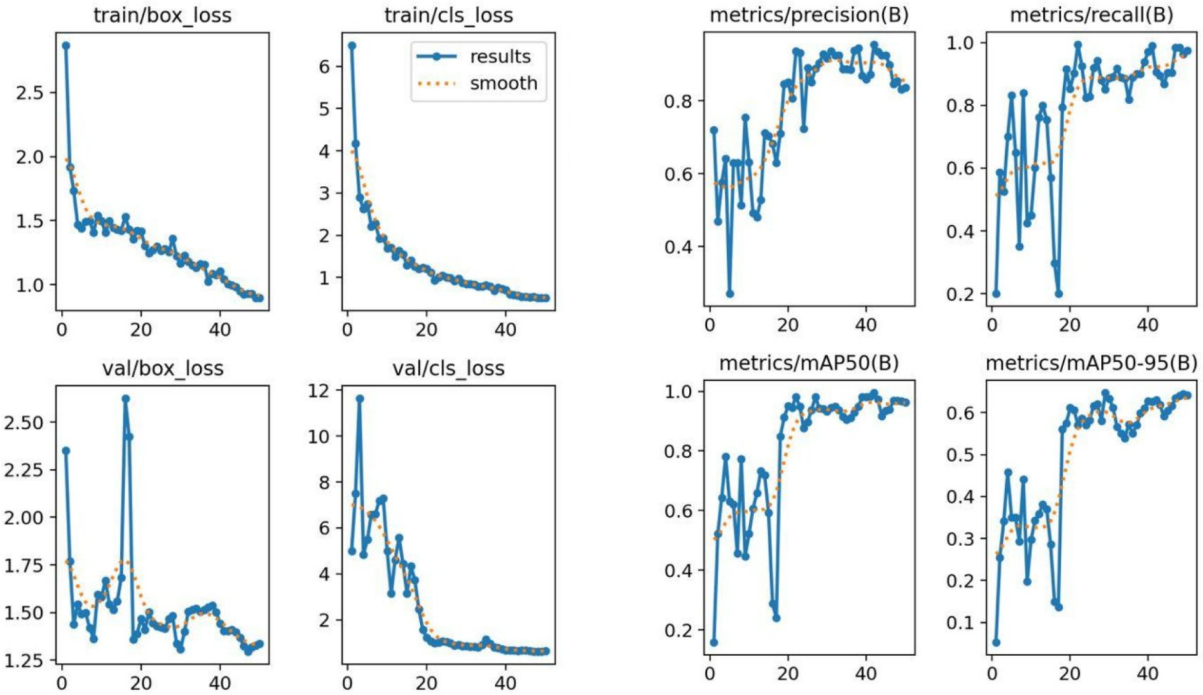




YOLOv8 Implementation

YOLOv8 models performs well in terms of precisely locating the ISL gestures , but when it comes to the precision calculated across a range of **IoU thresholds from 50% to 95%**, it indicates that the model is facing challenges in maintaining consister precision across a wider range of IoU thresholds.

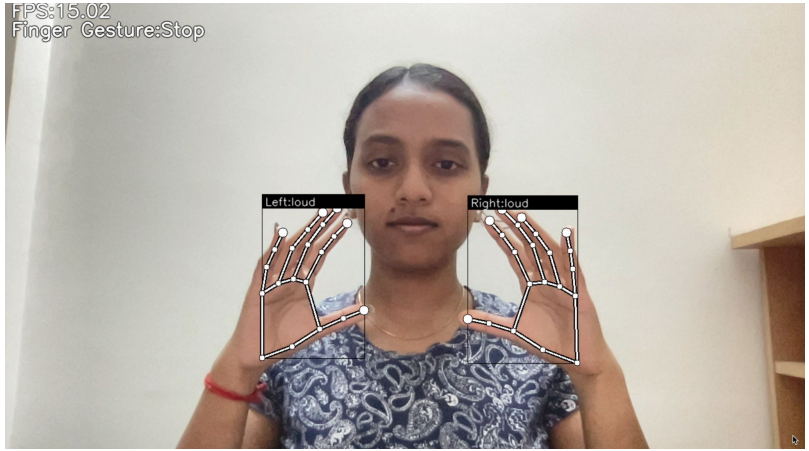
- mAP50 score value 0.84
- mAP50-95 score value 0.647



YOLOv8 Implementation

Class	Precision	Recall	mAP50	mAP50-95
all	0.829	0.852	0.84	0.647
alive	0.878	0.902	0.767	0.6
dead	0.737	1	0.895	0.717
fast	0.9	0.607	0.788	0.679
healthy	0.815	0.75	0.556	0.479
loud	0.873	0.71	0.695	0.563

CNN+RNN with Mediapipe Implementation

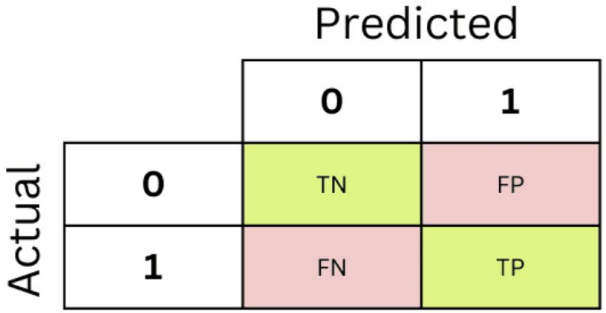




7 classes namely:

Loud, Mother, Healthy, Sick, Alive, Dead, Fast and Careful.

CNN+RNN with Mediapipe Implementation



$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

CNN+RNN with Mediapipe Implementation

This model exhibits robustness in classifying ISL gestures into 10 distinct categories. The high precision, recall and f1-score scores proves that this method has been the most precise for detection of hand gestures.

- Overall accuracy score 0.97
- Overall loss score value 0.1

Class	Precision	Recall	F1 score
alive	0.929	0.902	0.915
dead	0.98	1	0.989
fast	0.987	0.9	0.941
healthy	0.9	0.96	0.93
loud	0.86	0.892	0.875
mother	0.94	0.913	0.926
sick	0.955	0.89	0.921

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Thank you !